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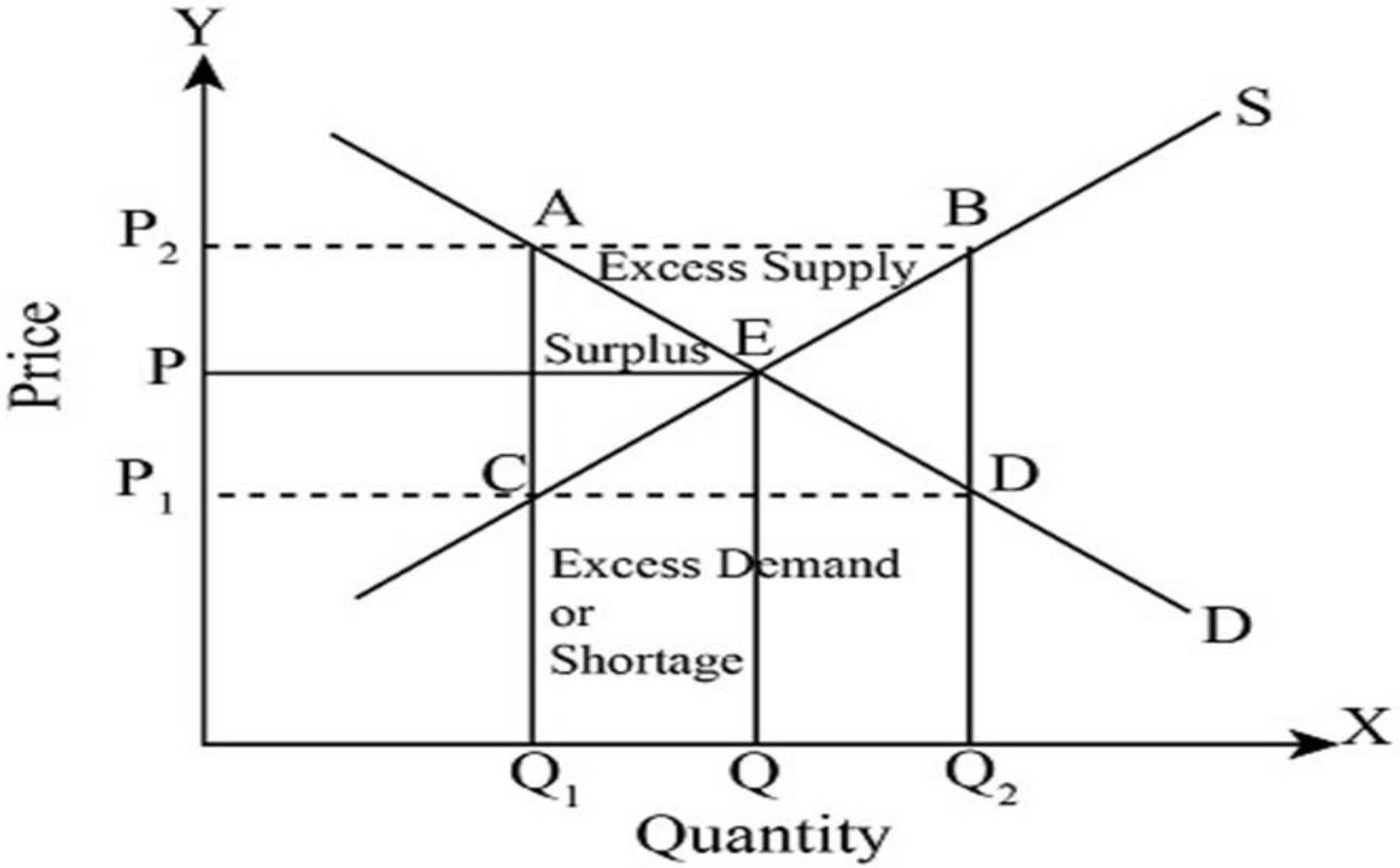
Department of Agriculture and Aquatic Sciences

Agribusiness Management (SA: 500)

Lecture 3: Economics for Agribusiness

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Lecture 3: Economics for Agribusiness



Introduction

- Agribusiness firms operate within a broader economic environment that strongly influences managerial decision-making.
- To be effective decision-makers, agribusiness managers must understand economics.
- Economics examines how individuals, firms, and societies make choices in the presence of scarcity.
- Economic choices directly affect production, marketing, investment, and profitability.

Objectives

After finishing this lecture, students should be able to:

- Explain the differences between accounting profit and economic profit
- Understand how supply and demand interact to determine market equilibrium

Introduction to Economics in Agribusiness

Definition of Economics

- Economics studies how individuals, firms, and society choose to best combine scarce resources (land, labor, capital, and management) to satisfy unlimited wants and needs of consumer.
- To meet unlimited human needs, economists need factors of production:

Factor	Reward
Land	Rent
Labor	Wage
Capital	Interest
Management	Salary/Profit

Introduction to Economics in Agribusiness

Scarcity

- Scarcity is the central economics problem faced by both producers and consumers.
- It means resources are limited (scarce) while human wants are unlimited. Thus:
- Scarcity forces **trade-offs and opportunity cost** in production and consumption.
- **Opportunity Cost** is the value of a resource in its next-best use. It is the next best alternative an individual or firm must give up in order to do something.
- Fundamental economic problem arises due to s**carcity thereby forcing choices**.
- Economics is about allocating scarce resources among competing ends.

Branches of Economics

There are basically two main branches of economics; (i) microeconomics and (i) macroeconomics.

(ii) Macroeconomics

- Macroeconomics focuses on the “big picture” view of the economic system.

It studies:

- National income
- Gross Domestic Product (GDP)
- Inflation
- Unemployment
- Interest rates

Branches of Economics

(i) Macroeconomics

Relevance to Agribusiness

- The rise in interest rates in Zambia can increase the cost of loan repayment leading to increased cost of production.
- During economic boom periods, demand for premium beef cuts increases.
- During recession, consumers shift to cheaper staple foods like mealie meal.

Branches of Economics

(ii) Microeconomics

- Microeconomics is the application of basic economic principles to decisions within the firm.

Agribusiness managers use microeconomics to:

- Allocate limited resources
- Decide production levels
- Set prices
- Maximize profit

Profit in Agribusiness

(i) Accounting Profit

- Accounting profit = Total Revenue – Explicit Costs
- Explicit costs = actual, measurable payments (wages, fuel, fertilizer, rent, utilities)

(ii) Economic Profit

- Economic profit = Accounting profit – Implicit Costs
- Implicit costs = opportunity costs of owned resources

Profit in Agribusiness

Types of Costs

(a) Explicit Costs

- Direct money payments.
- Examples: wages, rent, materials, utilities.
- Measured by accountants.

(b) Implicit Costs (Opportunity Costs)

- Value of best alternative use of resources.
- No cash payment made.
- Considered by economists.

Profit in Agribusiness

Formulae

- $\textbf{Accounting } \boldsymbol{Profit} = Total \text{ Revenue} - ExplicitCosts$
- $\textbf{Economic } \boldsymbol{Profit} = Total\ Revenue - (Explicit + ImplicitCosts)$

Why Does Economic Profit Exists?

- Reward for risk-taking
- Control of scarce resources
- Access to special information
- Superior management

Profit in Agribusiness: Scenario

Suppose a farmer earns:

- Revenue = K200,000
- Explicit costs (seed, fertiliser, labour) = K150,000

How much would be his/her profit?

- The profit = K50,000.
- **This is called accounting Profit = K50,000**

Suppose the farmer could have:

- Rented the land for K20,000
- Worked elsewhere and earned K25,000

How much is the real profit now?

Economic Profit = 50,000 – 45,000 = K5,000

Is this business attractive?

Would you continue farming?

You see why many farmers think they are profitable, but are actually not?

Profit in Agribusiness

Class Practice

Jerita owns and operates a landscaping business. She invested **K400,000** of her own money in the business and draws an annual salary of **K35,000**. She operates from a building owned by her parents, who allow her to use it rent-free, although it could be rented out for **K25,000** per year.

In a typical year, the business has total revenue of **K335,000** and the following costs:

- Plants and materials = K150,000
- Labor = K40,000
- Utilities = K25,000
- Taxes and insurance = K10,000
- Draw of salary = K35,000

Jerita could earn **K50,000** per year working for another company. Alternatively, she could invest her K400,000 in a mutual fund earning 8% annually.

Profit in Agribusiness

Typical Year Scenario

(a) Calculate Jerita's accounting profit. (5 marks)

(b) Calculate Jerita's total implicit costs. (5 marks)

(c) Calculate Jerita's economic profit. (5 marks)

(d) Explain whether Jerita should continue operating her business in the typical year. (5 marks)

(Bad Year Scenario)

In a bad year, accounting profit falls to **K60,000**, and Jerita receives a job offer paying **K65,000** per year.

(e) Calculate Jerita's economic profit in the bad year. (5 marks)

(f) Advise Jerita whether she should continue in business. Give reasons. (5 marks)

The Economics of Markets

WHAT IS A MARKET?

- A Market is the interaction between buyers and sellers.
- A market does not have to be a physical place.

Marketplace vs Market

- Marketplace is a physical location where buyers and sellers meet (e.g., Soweto Market).
- Market is interaction between buyers and sellers.

The Economics of Markets

Characteristics of a Market

There must be:

- A commodity which is dealt with.
- The existence of buyers and sellers.
- A place (region, country, or world).
- Free intercourse between buyers and sellers such that only one price should prevail for the same commodity at the same time.

The Economics of Markets

Market Structure

- Structure refers to organization, dimension, shape, size and design of a market.

Market structure determines:

- Level of competition.
- Pricing power.
- Profit opportunities.

Market structures are broadly classified as:

- Perfect Market (Perfect/Pure Competition)
- Imperfect Markets (Lack of competition)

The Economics of Markets

(i) Perfect Market (Perfect Competition)

- A perfectly competitive market is one with many buyers and sellers where no single participant can influence the market price.

Characteristics of perfect markets are as follows:

- There is a large number of buyers and sellers.
- All buyers and sellers have perfect knowledge of demand, supply and prices.
- Prices at any one time are uniform over a geographical area.
- Prices are uniform at any one place.
- Prices of different forms of a product are uniform.

Examples: Smallholder maize, groundnuts, soybeans sold in open markets.

The Economics of Markets

(ii) Imperfect Markets

- Markets where conditions of perfect competition are lacking.

Imperfect Markets include:

- Monopoly Market: For example, **ZESCO** as the sole electricity supplier to farms in Zambia.
- Duopoly Market: Only two sellers of a commodity. Not common in agribusiness
- Oligopoly Market: More than two, but still few large firms dominate the market. **E.g. MTN and Airtel** are good examples of oligopoly.
- Monopolistic Competition: A large number of sellers deal in heterogeneous and differentiated forms of a commodity. E.g. Different brands of fertilizers, processed meat products (Zambeef).

The Economics of Markets

Types of Market Structures

Structure	Number of Firms	Zambian Examples
Monopoly	Single seller	ZESCO (electricity distribution in many areas)
Oligopoly	Few sellers	Mobile network operators (MTN, Airtel, Zamtel)
Monopolistic Competition	Many sellers of branded goods	Seed companies, Fertilizer companies, Meat processors
Perfect Competition	Numerous sellers	Small-scale maize farmers, tomato traders on Open Market

The Economics of Markets

In agribusiness, understanding market structure helps managers know:

- Whether they can influence price
- Whether they must accept market price
- Whether branding or cost reduction is more important
- For example, under perfect competition, farmers are price takers, i.e. higher input costs (fertiliser, fuel) cannot be passed on to buyers.

The Economics of Markets

Supply and Demand

Supply and Demand are the two most important building blocks of economics. They explain:

- How markets (perfect competition markets) allocate goods and resources
- How prices are determined
- How producers and consumers respond to price changes

In agribusiness, supply and demand explain, for example:

- Why maize prices rise after a drought
- Why Chicken prices increase during festive seasons
- Why fertilizer use increases when crop prices increase

The Economics of Markets

Supply

- Supply refers to the quantity sellers are willing and able to sell at different prices during a given time period.

Law of Supply

- There is a direct relationship between price and quantity supplied.

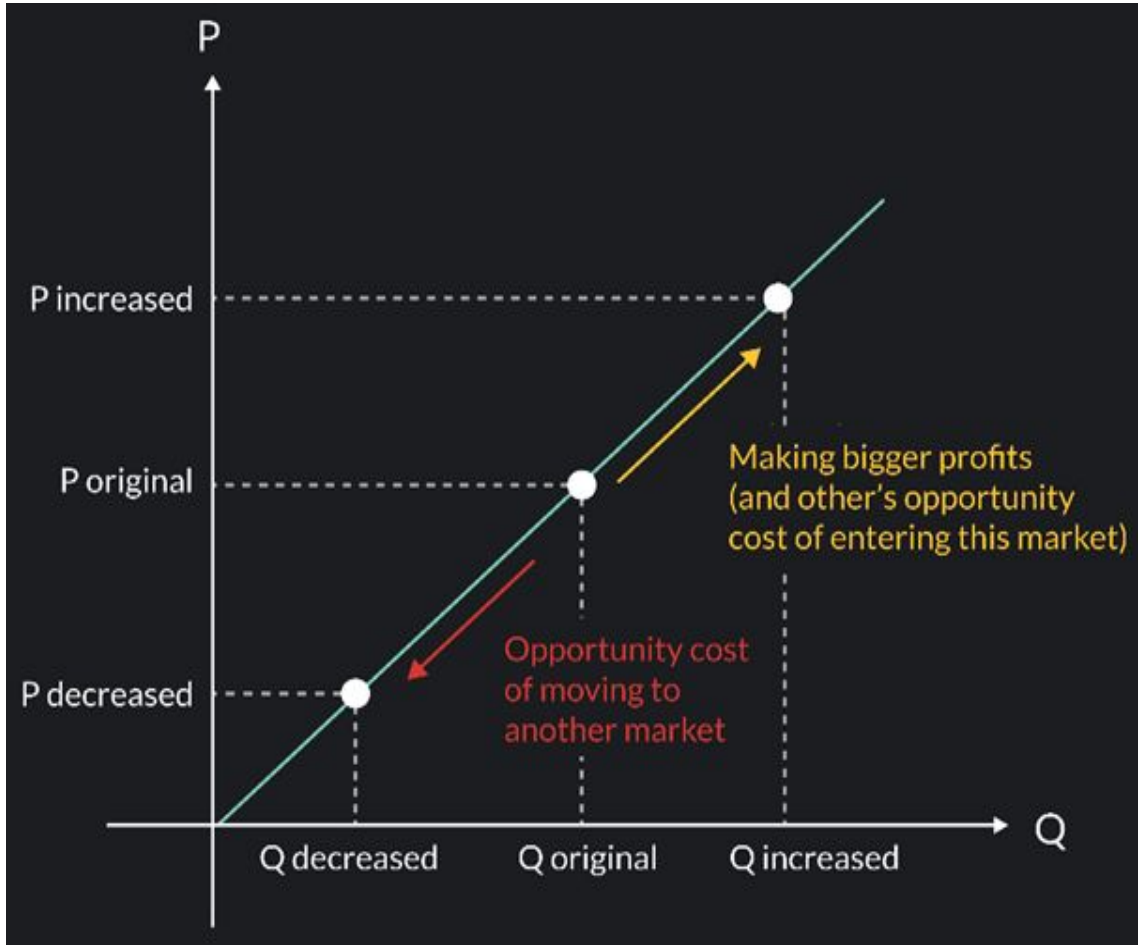
In functional form: $Q_s = f(P)$

Where:

- Q_s= Quantity supplied
- P= Price
- When price increases, quantity supplied also increases.

The Economics of Markets

Market Supply Curve



- When the price of a good increases, the quantity supplied increases, due to the profit-maximizing behaviour of individual firms.
- A market supply curve shows the positive relationship between the price and quantity supplied of a good.
- The Law of supply states that when the price of a good increases, the quantity supplied will also increase, holding all else constant.
- To maximize profits, individual firms will produce greater levels of output when prices are high.

The Economics of Markets

Demand

- Demand refers to the Quantity consumers are willing and able to buy at various prices during a given time period.

Law of Demand

- There is an inverse relationship between price and quantity demanded.

Demand depends on:

- $Q_d = f(P, P_{related}, M)$Where:
- P= own price
- $P_{related}$= price of related goods
- M= income

The Economics of Markets

Example: Student Food Budget

Assume:

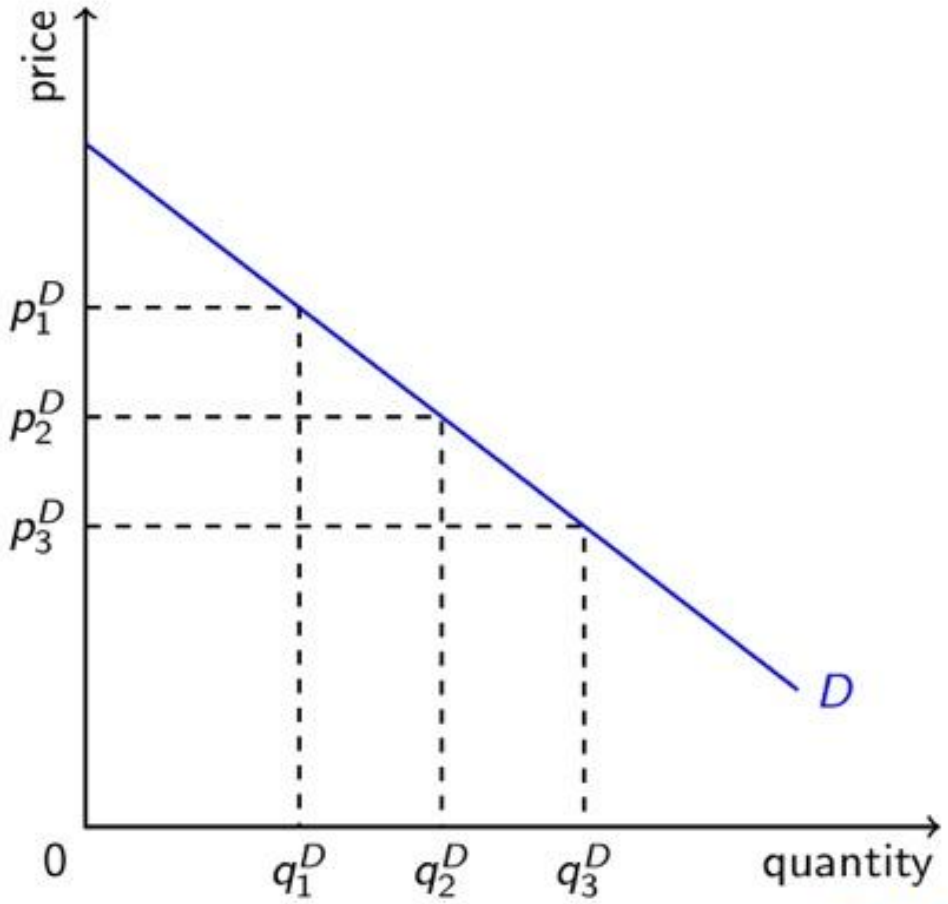
- Income (M) = K400 per week
- Macaroni price falls from K20 to K10

As price falls:

- Quantity demanded increases
- Purchasing power increases
- This creates a **demand schedule**.
- **Demand Schedule** is the information on prices and quantities purchased.

The Economics of Markets

Demand Curve



Interpretation

- Price on vertical axis (P)
- Quantity demanded on horizontal axis (Qd)
- Curve slopes downward

The Economics of Markets

Market Equilibrium

Markets bring together:

- Producers who want highest price
- Consumers who want lowest price
- The interaction leads to equilibrium.
- Equilibrium is a point from which there is no tendency to change.

The Economics of Markets

- **Market Equilibrium**

• The point where quantity supplied equals quantity demanded ($Q_s = Q_d$)

Equilibrium Price (P)

• The price where quantity supplied equals quantity demanded.

Also called:

• Market price

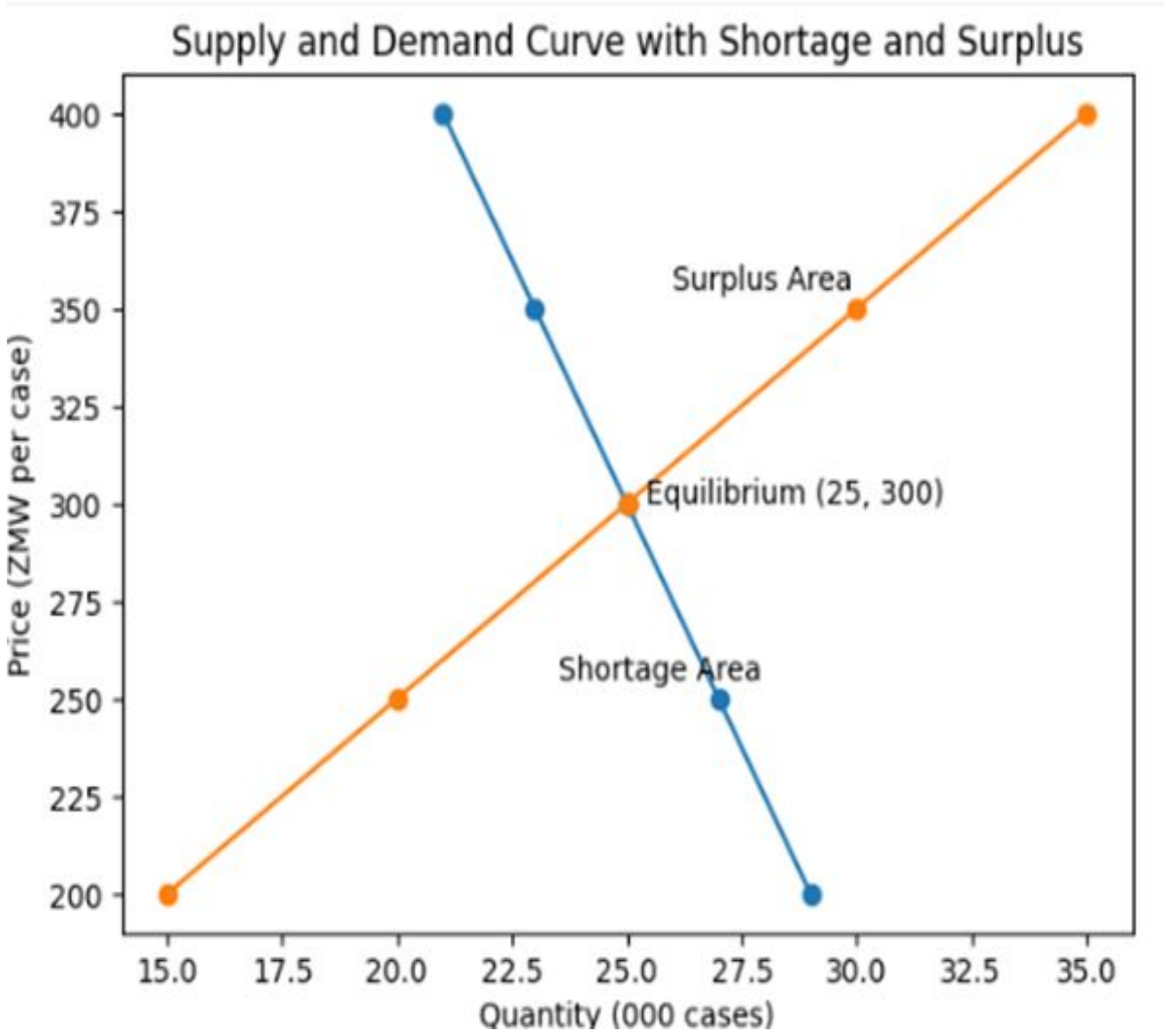
• Equilibrium Quantity (Q): The quantity bought and sold at equilibrium.

The Economics of Markets

Zambian Cereal Example

Supply and Demand Schedule

Price (ZMW per case)	Quantity Demanded (000)	Quantity Supplied (000)
200	29	15
250	27	20
300	25	25
350	23	30
400	21	35



The Economics of Markets

- Equilibrium at (Q = 25, P = ZMW 300)
- Shortage at ZMW 250
- **Surplus** at ZMW 350

Disequilibrium

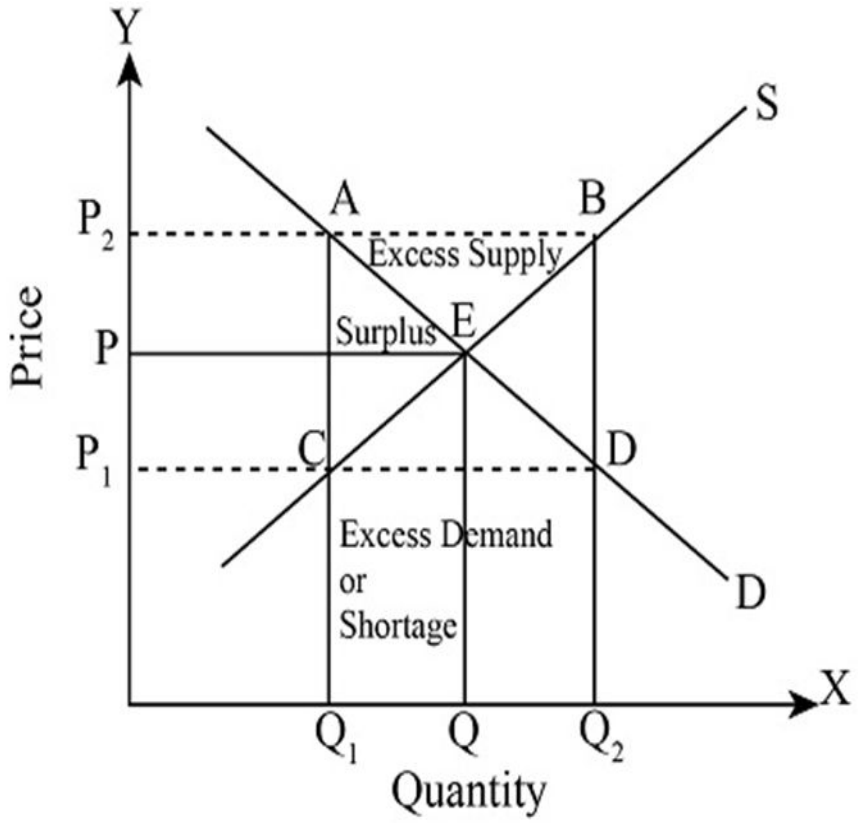
- Any price other than equilibrium creates imbalance.
- Disequilibrium is a situation where quantity supplied ≠ quantity demanded.

Two types of disequilibrium:

- Surplus
- Shortage

The Economics of Markets

SURPLUS



- Surplus is when quantity supplied is greater than quantity demanded (Qs > Qd).
- Surplus occurs when price is above equilibrium.

Example: Wheat Surplus

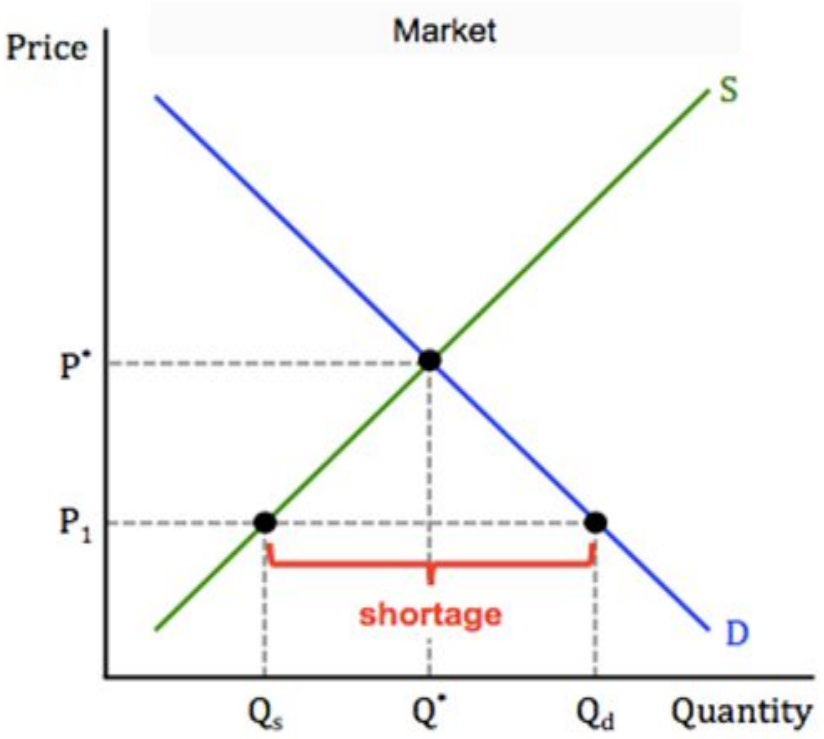
- If wheat price is too high:
- Farmers produce 50,000 tons
- Millers buy only 30,000 tons
- 20,000 tons remain unsold (Surplus).

What happens?

- Sellers reduce price.
- Surplus disappears.
- Price falls toward equilibrium.

The Economics of Markets

SHORTAGE



- Shortage is when quantity demanded is greater than quantity supplied (Qd > Qs).
- Shortage occurs when price is below equilibrium.

Example: Fertilizer Shortage

If government sets fertilizer price too low:

- Farmers demand 100,000 bags
- Suppliers provide only 60,000 bags
- Shortage of 40,000 bags.

What happens?

- Buyers compete.
- Price rises.
- Market returns to equilibrium.

The Economics of Markets

How Markets Self-correct

Situation	What Happens?
Surplus	Price falls
Shortage	Price rises
At equilibrium	No pressure to change

Markets are self-correcting institutions. Example:

If Shoprite has too much cooking oil:

- It lowers price.

If shelves are empty:

- It raises price and orders more stock.

Multiple Choice Questions for Recap

1. Economics is mainly concerned with:

A. Increasing profits only

B. Allocating scarce resources among competing ends

C. Production economics and its effect on profit maximization

D. Macroeconomic factors affecting agribusiness

Answer: B

Multiple Choice Questions for Recap

2. Scarcity means that:

A. Resources are unlimited

B. Human wants are limited

C. Resources are limited but wants are unlimited

D. Opportunity cost

Answer: C

Multiple Choice Questions for Recap

3. Opportunity cost is defined as:

A. The total revenue earned

B. The next best alternative forgone

C. The explicit cost of production

D. Economic profit

Answer: B

Multiple Choice Questions for Recap

4. Which of the following is an example of macroeconomics?

A. A farmer deciding how much maize to plant

B. Pricing vegetables at the Market

C. Studying inflation and unemployment in Zambia

D. Calculating fertilizer costs

Answer: C

Multiple Choice Questions for Recap

5. Accounting profit is calculated as:

A. Total Revenue – Implicit Costs

B. Total Revenue – Total Costs

C. Total Revenue – Explicit Costs

D. Explicit Costs – Implicit Costs

Answer: C

Multiple Choice Questions for Recap

6. Economic profit is different from accounting profit because it includes:

A. Explicit (opportunity) costs

B. Implicit (opportunity) costs

C. Total Revenue

D. Government subsidies

Answer: B

Multiple Choice Questions for Recap

7. A perfectly competitive market is characterized by:

A. One seller controlling production and prices

B. Two sellers

C. Many buyers and sellers with no control over price

D. Government price control

Answer: C

Multiple Choice Questions for Recap

8. ZESCO is an example of which market structure?

A. Perfect competition

B. Monopoly

C. Oligopoly

D. Monopolistic competition

Answer: B

Multiple Choice Questions for Recap

9. According to the law of supply:

A. Price and quantity supplied move in opposite directions

B. When price increases, quantity supplied increases

C. When price increases, quantity supplied decreases

D. Supply is always constant

Answer: B

Multiple Choice Questions for Recap

10. A shortage occurs when:

A. Qs > Qd

B. Qs = Qd

C. Qd > Qs

D. Price is above equilibrium

Answer: C

The End